

1 Model Implementation Plan

This section outlines the implementation plan for the multimodal choice model. The objective is to construct a series of model specifications that progressively incorporate additional behavioral features. Each specification will be estimated sequentially in order to evaluate improvements in model fit and interpretability.

1.1 Baseline Utility Structure

The utility that individual n associates with alternative i in choice task t and round r is defined as

$$U_{nit}^{(r)} = \mu_r V_{nit} + \varepsilon_{nit}^{(r)}$$

where

- μ_r is the round-specific scale parameter,
- V_{nit} is the deterministic utility component,
- $\varepsilon_{nit}^{(r)}$ follows an i.i.d. extreme value distribution.

For identification, the scale parameter of the first round is normalized:

$$\mu_1 = 1$$

Additional scale parameters μ_2 and μ_3 are estimated for later rounds.

1.2 Access and Egress Treatment

Access and egress segments are combined and assumed to share the same marginal utility parameters. Therefore, the generalized access/egress travel time is defined as

$$TT_{AE} = TT^{access} + TT^{egress}$$

This restriction reduces parameter dimensionality and assumes that travelers perceive first-mile and last-mile travel disutility similarly.

1.3 Bicycle Alternative Aggregation

Private bicycle and bike-share alternatives are merged into a single category:

$$Bicycle = \{bike, bikeshare\}$$

A single alternative-specific constant (ASC_{bike}) and associated parameters are estimated for the combined bicycle alternative.

1.4 Deterministic Utility Specification

The deterministic utility function is specified as

$$\begin{aligned} V_{nit} = & ASC_i + \beta_{AE,T} TT_{AE,nit} + \beta_{PT} TT_{nit}^{transit} \\ & + \beta_C Cost_{nit} + \beta_W Wait_{nit} + \beta_{TR} Transfer_{nit} \\ & + \beta_{SA} StagedAnchor_{nit} + \beta_{CX} Complexity_{nt} \\ & + \delta'(X_n \times M_{it}) \end{aligned}$$

where

- TT_{AE} is the combined access and egress travel time,
- $TT^{transit}$ is the in-vehicle transit travel time,
- $Cost$ denotes travel cost,
- $Wait$ denotes waiting time,
- $Transfer$ captures transfer disutility,
- $StagedAnchor$ represents the staged anchoring variable,
- $Complexity$ measures the complexity of the choice task,
- X_n represents socioeconomic attributes,
- M_{it} represents mode indicators.

1.5 Transfer Penalty Specifications

Two alternative specifications for transfer penalty will be tested.

1.5.1 Specification A: Binary Transfer Penalty

$$Transfer_{nit} = \begin{cases} 1 & \text{if a transfer occurs} \\ 0 & \text{otherwise} \end{cases}$$

This specification captures the presence of any transfer within the trip chain.

1.5.2 Specification B: Transfer Count

$$Transfer_{nit} = N_{transfer}$$

where $N_{transfer}$ represents the total number of transfers in the alternative.

1.6 Socioeconomic Interaction Effects

Socioeconomic variables are incorporated through interactions with mode indicators rather than as standalone variables. This approach allows the model to capture preference heterogeneity across traveler groups.

The following interaction terms will be considered:

$$\beta_{RH,inc} \cdot Income_n \times RideHailing_{it}$$

$$\beta_{RH,age} \cdot Age_n \times RideHailing_{it}$$

$$\beta_{Bike,age} \cdot Age_n \times Bicycle_{it}$$

These interactions capture potential differences in modal preferences across income and age groups.

1.7 Round-Specific Scale Parameters

To account for differences in response consistency across rounds, round-specific scale parameters will be estimated:

$$U_{nit}^{(r)} = \mu_r V_{nit} + \varepsilon_{nit}^{(r)}$$

where

$$\mu_1 = 1, \quad \mu_2 > 0, \quad \mu_3 > 0$$

This structure allows the model to capture possible fatigue, learning effects, or differences in task complexity across rounds.

1.8 Estimation Sequence

Model specifications will be implemented sequentially as follows:

1. Model A: Baseline LOS variables only
2. Model B: Add transfer penalty (binary)
3. Model C: Replace binary transfer penalty with transfer count
4. Model D: Add staged anchor variable
5. Model E: Add socioeconomic interaction terms
6. Model F: Add round-specific scale parameters
7. Model G: Add complexity variable and associated interactions

Model performance will be evaluated using log-likelihood, AIC, BIC, and parameter significance.